

What Makes a Successful Video Game?

A Data-Driven Analysis of Player Reception
and Commercial Performance

4

Datasets

64K+

Game Records

1.6M+

Reviews

Project Scope

Research Questions

- Which attributes (genre, platform, year) are most associated with high global sales?
- Is there a relationship between user/critic ratings and commercial success?
- Can a simple model classify games as successful or not?

Hypothesis

Games with higher user ratings and belonging to specific genres such as Action or RPG are more likely to achieve higher global sales.

Additionally, simple classification models can reasonably distinguish successful games from unsuccessful ones.

Data Sources

Video Game Sales 2024

PRIMARY

64,016 records · 14 columns

Core dataset for commercial success analysis — genre, platform, publisher, regional & global sales figures.

Metacritic Games & Reviews

SECONDARY

13,442 games · 1.6M reviews · 16 + 9 columns

Critic and user review scores. Used to examine the relationship between reception and sales performance.

Steam Games Dataset

SECONDARY

89,618 records · 47 columns

PC-specific metadata including tags, pricing, playtime and aggregated review sentiment for modern games.

Work Completed & Early Findings

Completed

- All 4 datasets loaded and inspected
- Dropped irrelevant columns (e.g. img)
- Missing value analysis — critic_score (~90%) and total_sales (~70%) heavily incomplete
- Merged Metacritic datasets via shared id column
- Initial EDA: genre sales, platform counts, metascore distribution, critic score vs sales

Key Findings

#1

Action

Top-selling genre by total global sales

~0.3

Correlation

Weak positive link between critic scores and sales

70–90

Metascore range

Where the vast majority of games fall

What's Next

01

Full Preprocessing

Handle missing values, encode categorical variables, define the binary success label using a sales threshold.

02

Deeper EDA

Sales trends over time, regional breakdowns, cross-dataset comparisons between Metacritic and Steam.

03

Machine Learning

Train and evaluate Logistic Regression and Decision Tree models for binary classification of game success.

04

Power BI Dashboards

Export cleaned data to Power BI to build interactive dashboards for the final presentation.